

BioSeq Explorer

Analysis Report

Generated	2026-06-12 16:14
Dataset	clean_dataset_A.csv
Filter variant	Variant A
HUBA report	Loaded
Total sequences	42
Gene sources	15
Flagged sequences	5

1. Quality Control — Summary Statistics

Metric	Mean	Median	Std	Min	Max	Q25	Q75
GC content	40.39%	36.67%	8.74%	23.59%	58.53%	35.86%	48.38%
N content	1.53%	0.00%	5.60%	0.00%	25.81%	0.00%	0.00%
Length (bp)	92146.2	60.0	328873.5	12.0	1554651.0	59.0	10649.0

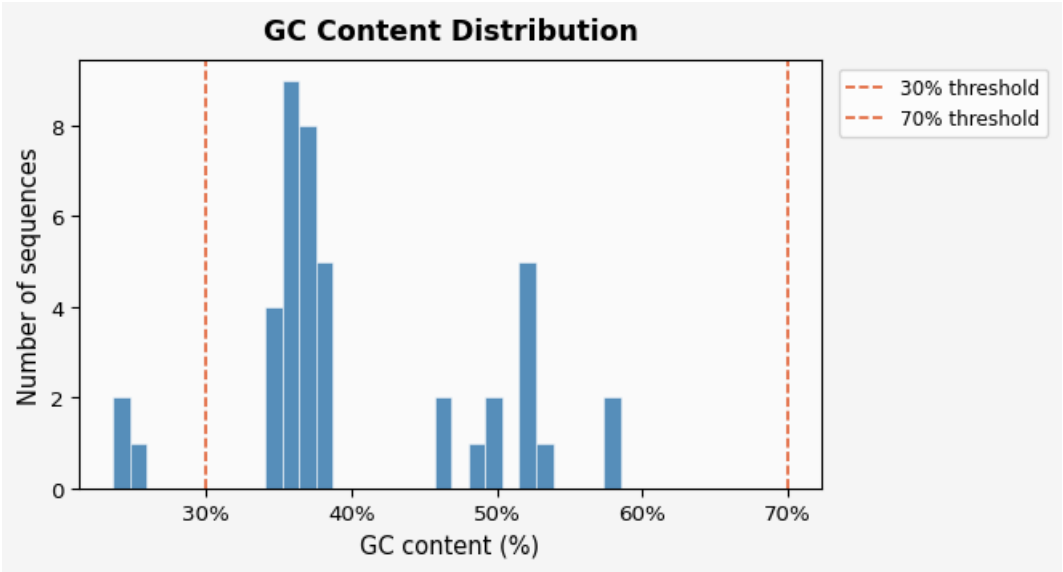
2. Per-Gene Statistics

Gene / Source	Mean GC%	Mean N%	Mean Length (bp)
brca1_sequences.fasta	36.69%	0.000%	10262.7
brca2_sequences.fasta	24.02%	0.000%	1244236.3
chk2_sequences.fasta	36.09%	0.000%	11874.7
institute_a_sequences.tsv	40.64%	0.000%	59.8
institute_b_sequences.tsv	39.95%	1.690%	59.5
institute_c_sequences.tsv	40.22%	0.000%	59.8
lab_a_sequences.csv	42.51%	0.000%	59.7
lab_b_sequences.csv	41.40%	0.000%	59.7
lab_c_sequences.csv	36.13%	3.390%	59.5
palb2_sequences.fasta	36.69%	0.000%	10262.7
sequences_batch1.fasta	42.69%	12.900%	30.5
sequences_batch2.fasta	50.00%	0.000%	26.0
sequences_batch3.fasta	43.75%	12.500%	34.0
test_invalid_chars.csv	50.00%	0.000%	12.0

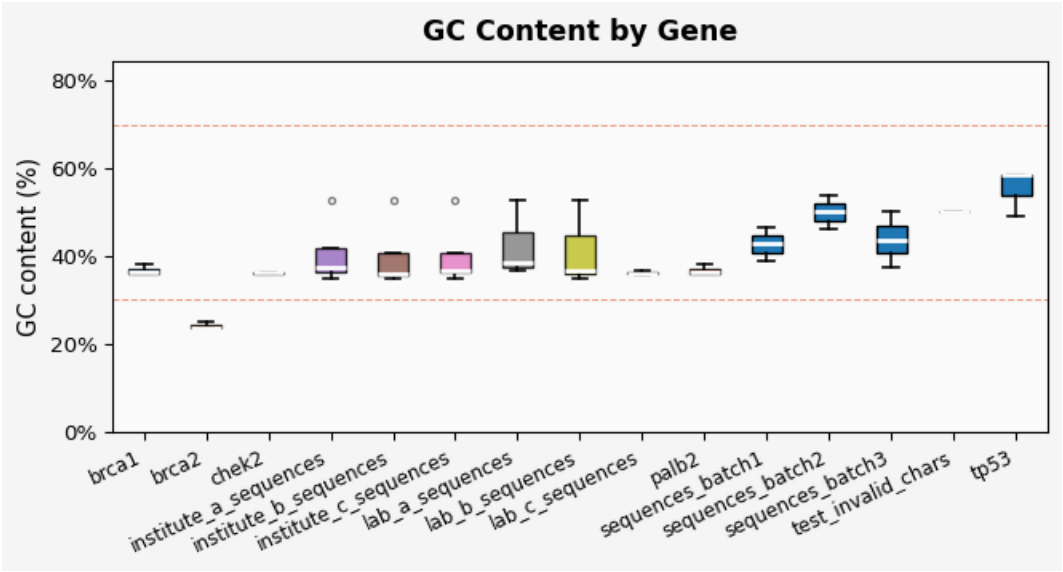
tp53_sequences.fasta	55.33%	0.000%	12948.3
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3. Visualizations

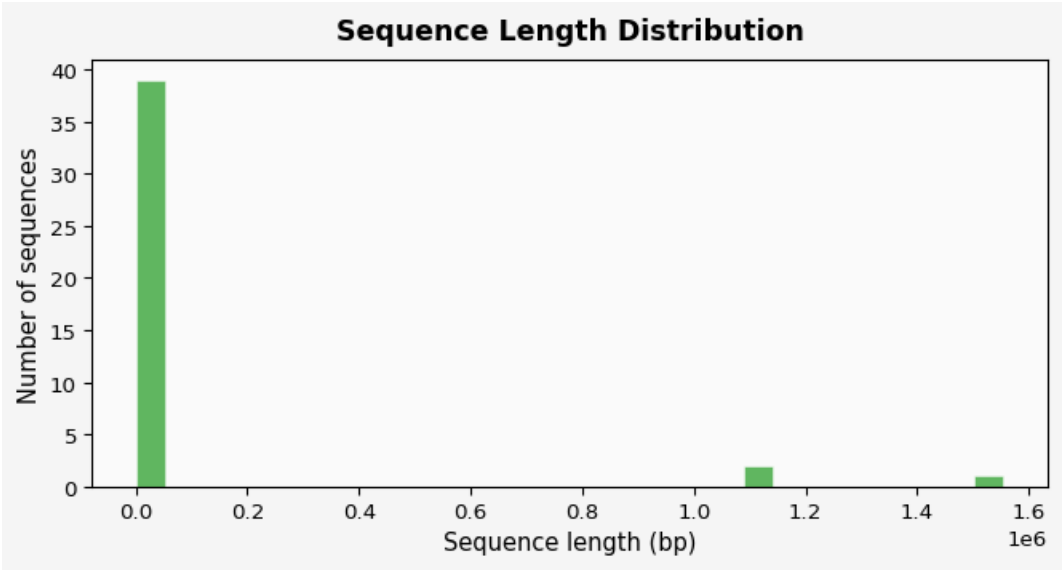
GC Content Distribution



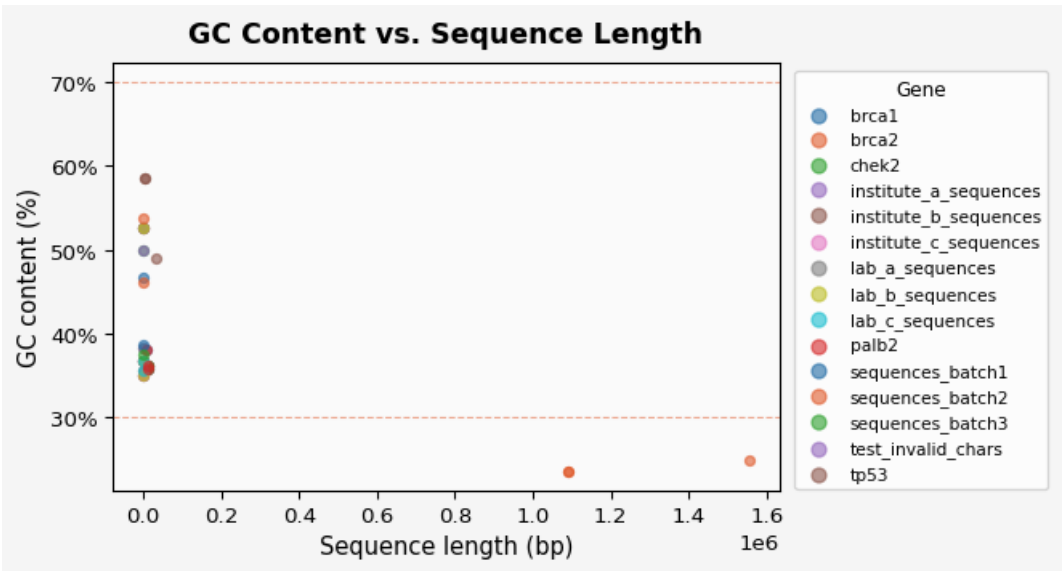
GC Content by Gene



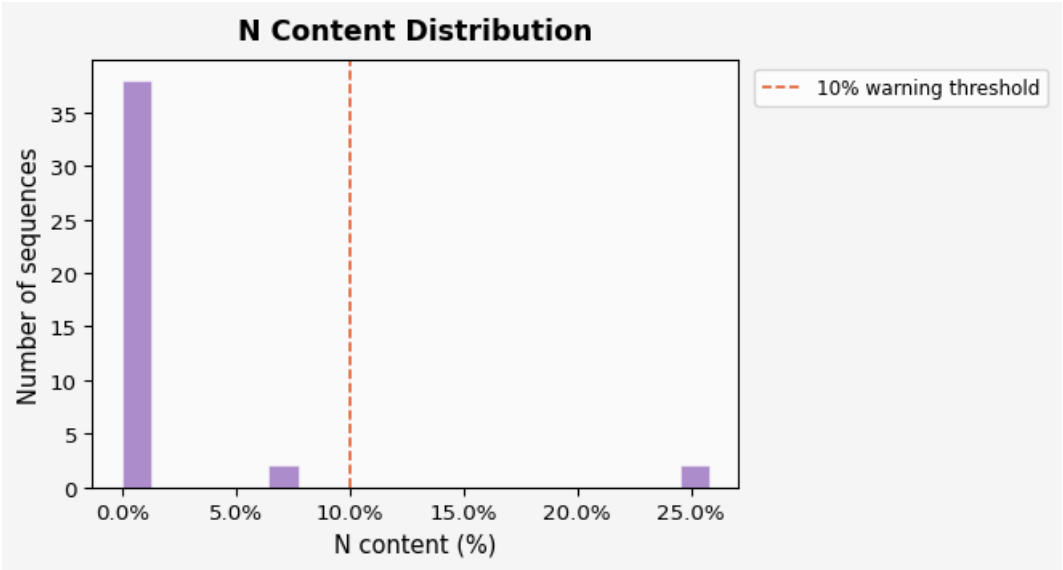
Sequence Length Distribution



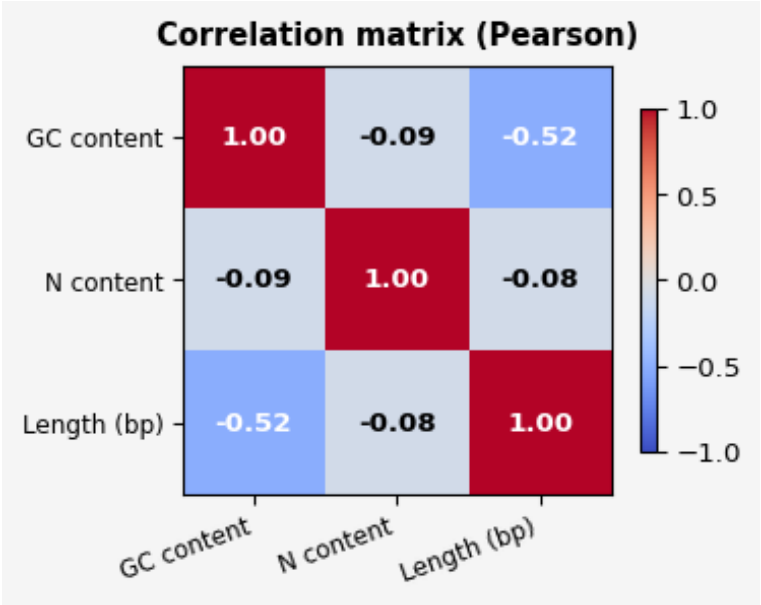
GC% vs. Sequence Length



N Content Distribution



Correlation Matrix



4. HUBA Pipeline Report

Data preparation summary generated by HUBA pipeline:

HUBA — Data Preparation Report

Input directory: source

Files loaded: 17

Input files

- `brca1\_sequences.fasta` (FASTA, 3 records)
- `brca2\_sequences.fasta` (FASTA, 3 records)
- `chek2\_sequences.fasta` (FASTA, 3 records)
- `institute\_a\_sequences.tsv` (TSV, 4 records)
- `institute\_b\_sequences.tsv` (TSV, 4 records)
- `institute\_c\_sequences.tsv` (TSV, 4 records)
- `lab\_a\_sequences.csv` (CSV, 3 records)
- `lab\_b\_sequences.csv` (CSV, 3 records)
- `lab\_c\_sequences.csv` (CSV, 3 records)
- `palb2\_sequences.fasta` (FASTA, 3 records)
- `sequences\_batch1.fasta` (FASTA, 3 records)
- `sequences\_batch2.fasta` (FASTA, 3 records)
- `sequences\_batch3.fasta` (FASTA, 2 records)
- `test\_empty.csv` (CSV, 2 records)
- `test\_invalid\_chars.csv` (CSV, 4 records)
- `test\_mixed\_case.csv` (CSV, 3 records)
- `tp53\_sequences.fasta` (FASTA, 3 records)

Variant A (min_len=10, max_n_pct=0.5)

- Accepted: 42/53
- Rejected: 11/53

Artefacts

- results/tables/file_profile.csv
- results/tables/input_stats.csv
- results/tables/clean_dataset_A.csv
- results/tables/after_filter_A.csv
- results/tables/clean_dataset_B.csv
- results/tables/after_filter_B.csv
- results/tables/clean_dataset_C.csv
- results/tables/after_filter_C.csv
- results/tables/rejected_A.csv
- results/tables/rejected_B.csv
- results/tables/rejected_C.csv
- results/tables/param_compare.csv
- results/plots/input_lengths.png
- results/plots/param_compare.png
- results/plots/gc_boxplot.png
- results/plots/gc_vs_length_scatter.png

Variant B (min_len=20, max_n_pct=0.2)

- Accepted: 39/53
- Rejected: 14/53

Variant C (min_len=50, max_n_pct=0.05)

- Accepted: 33/53
- Rejected: 20/53